**RAFFLES INSTITUTION**
**H2 Mathematics 9758**
**2026 Year 6 Term 3 Revision (Lambda)**
**Topic(s): (C) Statistics**
**(Solutions)**

**Source of Question: DHS Prelim 9758/2021/02/Q9**

**1** A manufacturer produces sweets in 2 flavours, lime and mint. On average, a proportion $p$ of the sweets are mint. The sweets are packed randomly in bags of 12 and the random variable $X$ denotes the number of mint sweets in each bag.

**(i)** State, in context, two assumptions needed for the number of mint sweets in a bag to be well-modelled by a binomial distribution. [2]

Assume now that the number of mint sweets in a bag has a binomial distribution.

**(ii)** Explain why the variance of $X$ does not exceed 3. [1]

**(iii)** Find the range of values of $p$ such that in a bag, the probability that there are 5 mint sweets is less than 0.1. Give your answer to 2 decimal places. [2]

**Take $p$ = 0.65 for the rest of the question.**

**(iv)** Find the probability that in a randomly selected bag, at least half of the sweets are mint. [2]

The bags are now randomly packed into boxes. Each box contains 5 bags.

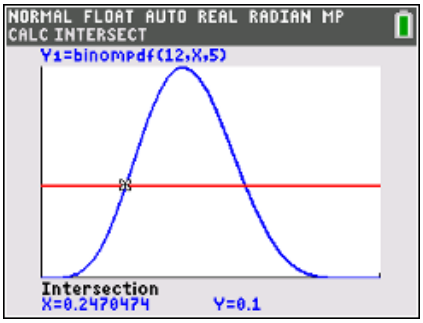
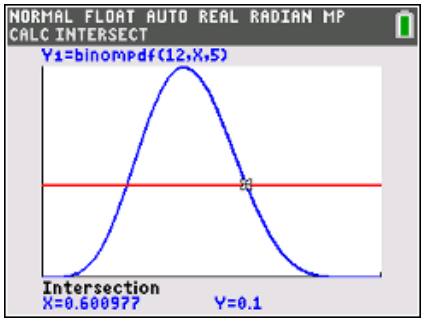
**(v)** Find the probability that in each of the bags in a randomly selected box, at least half of the sweets are mint. [1]

**(vi)** Find the probability that there are at least 30 mint sweets in a randomly selected box. [2]

**(vii)** Explain why the answer to part **(vi)** is greater than that in **(v)**. [1]

**Solution:**

| | |
|---|---|
| **(i)** | 1. The event that a sweet in the bag is mint flavoured is independent of another sweet being mint flavoured.<br>2. The probability of a randomly chosen sweet being mint is a constant $p$ for each sweet in a bag. |
| **(ii)** | Let $X$ denote the number of mint sweets in a bag of 12.<br>$X \sim \mathrm{B}(12, p)$<br>$\mathrm{Var}(X) = 12p(1-p) = -12(p-\frac{1}{2})^2 + 3$<br>Since $(p-\frac{1}{2})^2 \geq 0$,<br>$-12(p-\frac{1}{2})^2 \leq 0$<br>$-12(p-\frac{1}{2})^2 + 3 \leq 3$<br>$\therefore \mathrm{Var}(X) \leq 3$ |

| | |
|---|---|
| **(iii)** | $P(X=5)<0.1$<br>  <br>  <br>From GC, $p<0.24704$ or $p>0.60097$<br>$\therefore 0<p<0.25$ or $0.60<p<1$ (2 d.p.) |
| **(iv)** | $X \sim \mathrm{B}(12, 0.65)$<br>Required probability $= P(X \geq 6) = 1 - P(X \leq 5) = 0.91537 = 0.915$ (3 s.f.) |
| **(v)** | Required probability $= (0.91537)^5 = 0.643$ (3 s.f.) |
| **(vi)** | Let $Y$ denote the number of mint sweets in a box of 60 sweets.<br>$Y \sim \mathrm{B}(60, 0.65)$<br>Required probability $= P(Y \geq 30) = 1 - P(Y \leq 29) = 0.994$ (3 s.f.) |
| **(vii)** | The event in part **(v)** is a subset of the event in part **(vi)** and part **(vi)** contain cases which are not in part **(v)**, such as 3 bags having 4 mint sweets each and 2 bags having 9 mint sweets each. |

**Source of Question: ACJC Prelim 9758/2023/02/Q6**

**2** A factory manufactures a large number of chips in a variety of colours and packs them into boxes. Each box contains 36 randomly chosen chips. On average, 30% of the chips are yellow. You may assume that the number of yellow chips in a box follows a binomial distribution.

**(i)** A randomly chosen box is found to have at most 9 yellow chips. Find the probability that it contains more than 4 yellow chips. [2]

75 randomly chosen boxes are packed into a carton for shipment.

**(ii)** Find the probability that the $60^{\text{th}}$ box chosen is the $14^{\text{th}}$ box with at most 9 yellow chips. [2]

**(iii)** 30 cartons are sold to an arcade. Using a suitable approximation, find the probability that the mean number of boxes containing at most 9 yellow chips in each carton is more than 25. [3]

**Solution:**

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|---|---|
| **(i)** | Let $Y$ denote the number of yellow chips, out of 36.<br>$Y \sim \mathrm{B}(36, 0.3)$<br>$\mathrm{P}(Y \le 9) = 0.32544$<br>$\mathrm{P}(Y > 4 \mid Y \le 9) = \dfrac{\mathrm{P}(5 \le Y \le 9)}{\mathrm{P}(Y \le 9)}$<br>$= \dfrac{\mathrm{P}(Y \le 9) - \mathrm{P}(Y \le 4)}{\mathrm{P}(Y \le 9)}$<br>$= 0.97815 = 0.978$ (3s.f.) |
| **(ii)** | Let $W$ denote the number of boxes with at most 9 yellow chips, out of 59.<br>$W \sim \mathrm{B}(59, 0.32544)$<br>$\mathrm{P}(W = 13)$ = The probability that there are 13 boxes with at most 9 yellow chips each in the first 59 boxes.<br>So the probability that the 14th box with at most 9 yellow chips is the 60th box<br>$= \mathrm{P}(W = 13) \times \mathrm{P}(Y \le 9) = \mathrm{P}(W = 13) \times 0.32544 = 0.00825$ (3s.f) |
| **(iii)** | Let $X$ denote the number of boxes containing at most 9 yellow chips, out of 75.<br>$X \sim \mathrm{B}(75, 0.32544)$<br>$\mathrm{E}(X) = 75 \times 0.32544 = 24.408$<br>$\mathrm{Var}(X) = 75 \times 0.32544 \times (1 - 0.32544) = 16.465$<br>Since $n = 30$ is large, $\overline{X} \sim \mathrm{N}(24.408, \dfrac{16.465}{30})$ approximately by Central Limit Theorem.<br>$\mathrm{P}(\overline{X} > 25) = 0.21212 = 0.212$ (3 s.f.) |

**Source of Question: DHS Prelim 9758/2021/02/Q9**

**3** Emma has a computer program that generates a random positive integer $X$. The probability distribution of $X$ is:

$$\mathrm{P}(X=r)=\frac{a}{r^3},\quad r\in\mathbb{Z}^+ \text{ and } a \text{ is positive constant.}$$

For the rest of the question, you may use the following results:

$$\sum_{m=1}^{\infty}\frac{1}{m} \text{ does not exist, } \sum_{m=1}^{\infty}\frac{1}{m^2}=1.6449 \text{ and } \sum_{m=1}^{\infty}\frac{1}{m^3}=1.2021.$$

**(a)** Find the value of $a$. [2]

**(b)** Find $\mathrm{E}(X)$ and explain why $\mathrm{Var}(X)$ cannot be calculated. [2]

**(c)** Find $\mathrm{P}(X\geq 2\,|\,X\leq 15)$. [3]

Emma generates 10 numbers using her program namely, $X_1, X_2, \ldots, X_{10}$. The random variable $Y$ denotes the number of times the number '3' occurs among the 10 numbers. You can assume that the numbers generated by the program are independent of each other.

**(d)** Find $\mathrm{P}(Y>2)$. [2]

**(e)** Find $\mathrm{P}(X_1+Y=3)$. [4]

**Solution:**

**(a)**

$$\sum_{r=1}^{\infty}\mathrm{P}(X=r)=1$$

$$\sum_{r=1}^{\infty}\frac{a}{r^3}=1\Rightarrow a\sum_{r=1}^{\infty}\frac{1}{r^3}=1$$

$$a=\frac{1}{1.2021}=0.83188=0.832 \text{ (3 s.f.)}$$

**(b)**

$$\mathrm{E}(X)=\sum_{r=1}^{\infty}r\,\mathrm{P}(X=r)=a\sum_{r=1}^{\infty}\frac{1}{r^2}=(0.83188)(1.6449)=1.37 \text{ (3 s.f.)}$$

$$\mathrm{E}(X^2)=\sum_{r=1}^{\infty}r^2\,\mathrm{P}(X=r)=a\sum_{r=1}^{\infty}\frac{1}{r} \text{ does not exist.}$$

Therefore $\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2$ cannot be calculated.

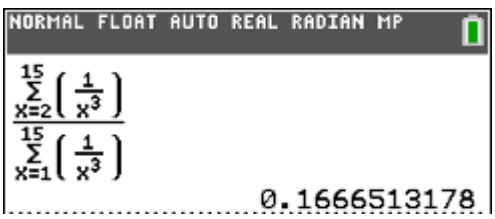
**(c)**

$$\mathrm{P}(X\geq 2\,|\,X\leq 15)$$

$$=\frac{\mathrm{P}(2\leq X\leq 15)}{\mathrm{P}(X\leq 15)}$$

$$=\frac{\sum_{r=2}^{15}\frac{a}{r^3}}{\sum_{r=1}^{15}\frac{a}{r^3}}=\frac{a\sum_{r=2}^{15}\frac{1}{r^3}}{a\sum_{r=1}^{15}\frac{1}{r^3}}$$

$$=0.16665=0.167 \text{ (3 s.f)}$$



| | |
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| **(d)** | $Y \sim \text{B}(10, \text{P}(X=3))$ where $\text{P}(X=3) = \frac{a}{3^3} = \frac{0.83188}{27} = 0.030810$<br>$Y \sim \text{B}(10, 0.030810)$<br>$\text{P}(Y>2) = 1 - \text{P}(Y \le 2) = 0.00298 \, (3 \text{ s.f.})$ |
| **(e)** | We consider the various cases for $X_1$ and $Y$.<br>$\text{P}(X_1 + Y = 3) = \text{P}(X_1 = 1, Y = 2) + \text{P}(X_1 = 2, Y = 1) + \text{P}(X_1 = 3, Y = 0)$<br>Note: $X_1$ and $Y$ are not independent.<br><br>**Case 1:** $X_1 = 1$ and $Y = 2$<br>The first number must be a '1' and the rest of the 9 numbers must have two '3's.<br>$\text{P}(X=1)\left[\binom{9}{2}(\text{P}(X=3))^2(1-\text{P}(X=3))^7\right] = 0.022835$<br><br>**Case 2:** $X_1 = 2$ and $Y = 1$<br>The first number must be a '2' and the rest of the 9 numbers must have one '3'.<br>$\text{P}(X=2)\left[\binom{9}{1}(\text{P}(X=3))(1-\text{P}(X=3))^8\right] = 0.022448$<br>**Note:** $X_1 = 3$ and $Y = 0$<br>This case is impossible as $Y$ counts the number of '3' generated, so since $X_1 = 3$, probability is 0 for this case.<br>$\therefore \text{P}(X_1 + Y = 3) = 0.022835 + 0.022448 + 0 = 0.0453$ (3 s.f.) |

**Source of Question: EJC Prelim 9758/2021/02/Q10(i)-(iv)**

**4** In this question, you should state the parameters of any distributions that you use.

A stationery factory manufactures pens for sale. The diameters (in mm) of the pens have distribution $N(10, 0.003)$. The pens are packed in sets of 24 into a box. Within the box, the pens are laid out side by side. The widths of the boxes (in mm) are normally distributed with mean 240.5 mm and variance 0.02 mm$^2$.

**(i)** Find the probability that a random sample of 24 pens would fit into a randomly selected box. [3]

A pen is considered "defective" if its diameter is not within 0.1mm of the population mean.

**(ii)** Find the probability that a randomly selected pen is defective. [2]

Pens are produced and inspected in batches. For each batch, a sample of 12 pens is randomly selected and checked.

- If there are fewer than 2 defective pens in this sample of 12, the batch passes the inspection.

- If there are exactly 2 defective pens in this sample, a second sample of 12 pens is randomly selected and checked. If there are no defective pens in the second sample, the batch passes the inspection.
- Otherwise, the batch does not pass the inspection.

**(iii)** Show that the probability of a batch of pens passing the inspection is 0.871, correct to 3 significant figures. [3]

**(iv)** Find the probability that not more than 3 defective pens were found in a batch of pens during the inspection process, given that the batch of pens did not pass the inspection. [3]

**Solution:**

**(i)**

Let $W$ denote the width (in mm) of a pen. $W \sim N(10, 0.003)$

Let $Y$ denote the width (in mm) of a box. $Y \sim N(240.5, 0.02)$

Let $C = W_1 + W_2 + ... + W_{24} - Y$

$C \sim N(-0.5, 0.092)$

$$\begin{aligned} P(\text{pens fit into box}) &= P(W_1 + W_2 + ...W_{24} < Y) \\ &= P(C < 0) \\ &= 0.95037 \\ &= 0.950 \text{ (3 s.f.)} \end{aligned}$$

**(ii)**

$P(\text{pen is defective}) = 1 - P(9.9 < W < 10.1) = 0.067889 = 0.0679 \text{ (3 s.f.)}$

**(iii)**

Let $X$ denote the number of defective pens in a box of 12. $X \sim B(12, 0.067889)$

$$\begin{aligned} &P(\text{batch passes inspection}) \\ &= P(X \le 1) + P(X = 2)P(X = 0) \\ &= 0.80609 + (0.15060)(0.43014) \\ &= 0.87087 = 0.871 \text{ (3 s.f.)} \end{aligned}$$

**(iv)**

$$\begin{aligned} &P(\le 3 \text{ defective pens found} \mid \text{batch did not pass inspection}) \\ &= \frac{P(X = 2)P(X = 1) + P(X = 3)}{P(\text{batch did not pass inspection})} \\ &= \frac{(0.15060)(0.37595) + 0.036562}{1 - 0.87087} \\ &= 0.72160 = 0.722 \text{ (3 s.f.)} \end{aligned}$$

**Source of Question: DHS Prelim 9758/2023/02/Q9**

**5** In this question you should state the parameters of any distributions that you use.

In a supermarket, the masses in grams of Fuji apples have the distribution $\mathrm{N}\left(205, 9^2\right)$ and the masses in grams of Gala apples have the distribution $\mathrm{N}\left(180, 6^2\right)$.

**(a)** Find the probability that a randomly chosen Fuji apple has a greater mass than a randomly chosen Gala apple. [2]

**(b)** Three Fuji apples are selected at random. Find the probability that two Fuji apples have mass exceeding 203 grams each and one Fuji apple has mass less than 185 grams. [2]

**(c)** The supermarket also sells assorted packets of apples containing a mixture of a total of 10 Fuji and Gala apples. If there are $n$ Fuji apples in an assorted packet, find the least value of $n$ such that there is a probability of at least 0.5 that the mass of an assorted packet exceeds nine times the mass of a Fuji apple by more than 28 g. [5]

**(d)** State an assumption needed for your calculations in parts **(a)** and **(c)**. [1]

**Solution:**

**(a)**

Let $F$ and $G$ denote the mass of a Fuji and Gala apple respectively, in grams.

$F \sim \mathrm{N}\left(205, 9^2\right), \quad G \sim \mathrm{N}\left(180, 6^2\right)$

$F - G \sim \mathrm{N}\left(205 - 180,\ 9^2 + 6^2\right) \Rightarrow F - G \sim \mathrm{N}\left(25,\ 117\right)$

$\mathrm{P}\left(F > G\right) = \mathrm{P}\left(F - G > 0\right)$

$= 0.98960 = 0.990$ (3 s.f.)

**(b)**

Required probability

$= \left[\mathrm{P}\left(F > 203\right)\right]^2 \times \mathrm{P}\left(F < 185\right) \times \dfrac{3!}{2!}$

$= \left(0.58793\right)^2 \times 0.013134 \times 3$

$= 0.013620 = 0.0136$ (3 s.f.)

**(c)**

Let $A$ denote the mass of an assorted packet of ten apples, in grams.

$A = \left(F_1 + F_2 + \ldots + F_n\right) + \left(G_1 + G_2 + \ldots + G_{10-n}\right)$

$\mathrm{E}\left(A\right) = n\mathrm{E}\left(F\right) + \left(10 - n\right)\mathrm{E}\left(G\right) = 205n + \left(10 - n\right)180 = 25n + 1800$

$\mathrm{Var}\left(A\right) = n\mathrm{Var}\left(F\right) + \left(10 - n\right)\mathrm{Var}\left(G\right) = 9^2 n + \left(10 - n\right)\left(6^2\right) = 45n + 360$

$A \sim \mathrm{N}\left(25n + 1800,\ 45n + 360\right)$

$\therefore A - 9F \sim \mathrm{N}\left(25n + 1800 - 9(205),\ 45n + 360 + 9^2\left(9^2\right)\right)$

$\Rightarrow A - 9F \sim \mathrm{N}\left(25n - 45,\ 45n + 6921\right)$

$\mathrm{P}\left(A - 9F > 28\right) \geq 0.5$

**Method 1: Standardisation**

$$\mathrm{P}\left(Z > \frac{28-(25n-45)}{\sqrt{45n+6921}}\right) \geq 0.5$$

$$\Rightarrow \frac{28-(25n-45)}{\sqrt{45n+6921}} \leq 0$$

$$\Rightarrow 28-(25n-45) \leq 0$$

$$\Rightarrow n \geq 2.93$$

Least $n$ in an assorted packet is 3.

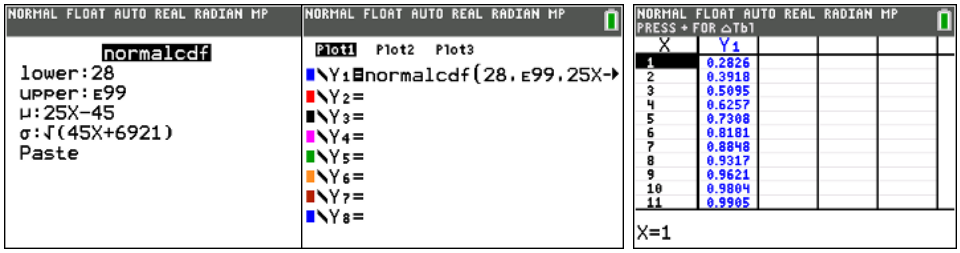
**Method 2: Graphic Calculator**

From GC, $\mathrm{P}(A-9F>28) \geq 0.5$

When $n=2$, $\mathrm{P}(A-9F>28) = 0.3918 < 0.5$

When $n=3$, $\mathrm{P}(A-9F>28) = 0.5095 > 0.5$

Least $n$ in an assorted packet is 3.




**(d)** The mass of **every** apple is independent of every other apple.

**Source of Question: ACJC Prelim 9758/2023/02/Q11**

**6** Energy bars are a convenient source of nutrition for athletes. A typical energy bar weighs between 30-50 g and is likely to supply about 200-300 calories. An American company sells energy bars and a nutritionist working in the company claims that the mean number of calories in a product, Perfect Protein bars, is 350 calories.

A random sample of 30 bars is collected and the number of calories, $x$ cal, in the 30 bars is summarised as follows.

$$\sum x = 10446 \text{ , } \sum x^2 = 3638000 \text{ .}$$

**(i)** Test at the 3% level of significance whether the mean number of calories in Perfect Protein bars is less than 350 calories, defining any symbols that you use. [7]

**(ii)** Explain why the test would be inadmissible if the nutritionist had taken a random sample of 15 energy bars and hence state an assumption needed for him to carry out the test. [1]

**(iii)** If the test was for whether the mean number of calories in Perfect Protein bars is 350 calories, find the smallest level of significance for which the nutritionist's claim will be rejected. [2]

The company launched a new product called Dynamic Protein bars which are advertised to be higher in calories than Perfect Protein bars. A random sample of 45 Dynamic Protein bars is collected to test this and the mean number of calories of the sample is found to be 352.2.

**(iv)** The result of this investigation shows that there is no reason to believe that the mean number of calories in the new product is more than 350 calories at the 4% level of significance. Find the range of possible values of the variance used in calculating the test statistic. [3]

**Solution:**

**(i)**

$\overline{x} = \dfrac{\sum x}{n} = \dfrac{10446}{30} = 348.2$ (exact)

$s^2 = \dfrac{1}{n-1}\left(\sum x^2 - \dfrac{\left(\sum x\right)^2}{n}\right) = \dfrac{1}{29}\left(3638000 - \dfrac{(10446)^2}{30}\right) = 24.234$

Let $\mu$ be the population mean number of calories in Perfect Protein bars.

Null hypothesis $H_o: \mu = 350$

Alternative hypothesis $H_1: \mu < 350$

Perform a 1-tail test at 3 % significance level.

Under $H_0$, $\overline{X} \sim N\left(350, \dfrac{24.234}{30}\right)$ approximately by Central Limit Theorem since $n = 30$ is large enough.

From the sample, $\overline{x} = 348.2$.

From GC, $p\text{-value} = P\left(\overline{X} \le 348.2\right) = 0.022604 = 0.0226$

Since $p$-value < 0.03, we reject $H_o$ and conclude that there is sufficient evidence at 3% level of significance that the mean number of calories in Perfect Protein bars is less than 350 cal.

**(ii)** The test would be inadmissible if the nutritionist had taken a random sample of 15 energy bars as the distribution of the population is unknown (and Central Limit Theorem cannot be applied to approximate the distribution of the sample mean number of calories since sample size of 15 is not large enough). The nutritionist will need to assume that the number of calories in Perfect Protein bars follow a normal distribution.

**(iii)** Null hypothesis $H_o: \mu = 350$

Alternative hypothesis $H_1: \mu \neq 350$

Perform a 2-tail test at $\alpha$ % significance level.

$p$-value $= 2 \times P\left(\overline{X} \leq 348.2\right) = 0.045208 = 0.0452$ (3 s.f.)

To reject $H_o$, $p$-value $= 0.0452 \leq \alpha\%$

$\therefore$ smallest significance level for which claim is rejected is 4.52%.

**(iv)** Let $\mu_y$ be the population mean number of calories in Dynamic Protein bars.

Null hypothesis $H_o: \mu_y = 350$

Alternative hypothesis $H_1: \mu_y > 350$

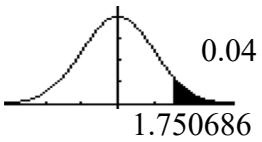
Under Ho, $\overline{Y} \sim N\left(350, \frac{\sigma^2}{45}\right)$ approximately by Central Limit Theorem since $n = 45$ is large enough.

At 4% level of significance, we do not reject $H_o$.

$\Rightarrow$ $p$-value $> 0.04$

That is, $P(\overline{Y} > 352.2) > 0.04$

$$P\left(Z > \frac{352.2 - 350}{\sqrt{\frac{\sigma^2}{45}}}\right) > 0.04$$




From GC, $P(Z > 1.75068) = 0.04$.

From diagram, $\frac{352.2 - 350}{\sqrt{\frac{\sigma^2}{45}}} < 1.750686$

$$1.750686\sqrt{\frac{\sigma^2}{45}} > 2.2$$

$$\sigma^2 > 45\left(\frac{2.2}{1.750686}\right)^2$$

$$\sigma^2 > 71.0626$$

$$\therefore \sigma^2 > 71.1 \text{ (3 s.f.)}$$

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